

4.1 EXPONENTS / INDICES**4.1.1 Identify base, exponent and value**

We know that if a number is repeatedly multiplied by itself then it can be represented in index or exponential notation.

For example, $2 \times 2 \times 2 \times 2 \times 2 = 2^5$

or $2^5 = 32$

Here 2 is base,

5 is exponent or index.

32 is the value which is the fifth power of 2.

and we read it as “2 to the power 5 is 32”.

Now, we define the terms related to index notation as:

Base:

Base is a number which is to be multiplied by itself according to exponent.

Exponent or Index:

Exponent is a number which represents the number of times, a base is to be multiplied by itself.

Value of exponential form:

The product of base with itself according to the exponent is the value of exponential form.

For example,

$$3^4 = 81 \quad \text{or} \quad 3 \times 3 \times 3 \times 3 = 81$$

Here 81 is the value of “3 to the power 4”

Example 1. Identify base and exponent in the following. Also find the value.

(i) 2^4 (ii) 5^2 (iii) $\left(\frac{4}{7}\right)^3$ (iv) $\left(\frac{2}{3}\right)^4$

Solution:

S. No.	Expression	Base	Exponent	Value
(i)	2^4	2	4	16 because $2 \times 2 \times 2 \times 2 = 16$
(ii)	5^2	5	2	25 because, $5 \times 5 = 25$
(iii)	$\left(\frac{4}{7}\right)^3$	$\frac{4}{7}$	3	$\frac{64}{343}$ because, $\frac{4}{7} \times \frac{4}{7} \times \frac{4}{7} = \frac{64}{343}$
(iv)	$\left(\frac{2}{3}\right)^4$	$\frac{2}{3}$	4	$\frac{16}{81}$ because, $\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{16}{81}$

EXERCISE 4.1

1. Expand and find the value of each exponential form.

- (i) 3^2 (ii) 2^7 (iii) $2^3 \times 3^4$
 (iv) $5^2 \times 7^2$ (v) $2^3 \times 3^2 \times 5^2$ (vi) $2^4 \times 3^3 \times 7^2$

2. Identify the base and exponent. Also find the value of each of the following expression.

- (i) 2^5 (ii) 5^4 (iii) 6^3 (iv) $\left(\frac{2}{3}\right)^6$ (v) $\left(-\frac{1}{5}\right)^5$

3. Show that:

- (i) $(-3)^3 = -27$ (ii) $\left(\frac{1}{2}\right)^7 = \frac{1}{128}$ (iii) $\left(-\frac{1}{4}\right)^4 = \frac{1}{256}$
 (iv) $\left(-\frac{1}{10}\right)^6 = \frac{1}{1000000}$ (v) $\left(-\frac{5}{3}\right)^4 = \frac{625}{81}$ (vi) $\left(\frac{4}{5}\right)^5 = \frac{1024}{3125}$

4. Express each rational number in exponential form.

- (i) 1000 (ii) 512 (iii) -343 (iv) 625
 (v) $-\frac{1}{100000}$ (vi) $\frac{32}{243}$ (vii) $-\frac{125}{343}$ (viii) $\frac{729}{64}$

4.2 LAWS OF EXPONENTS / INDICES

4.2.1 Use rational numbers to deduce laws of exponents.

There are some laws of exponents which help us to simplify expressions of exponential form. These laws are given below:

Product law.

- (I) **When bases are same but exponents are different, then product law is: $a^m \times a^n = a^{m+n}$ where a is a rational number and m, n are natural numbers.**

In order to deduce this law, let us consider the following examples.

$$\begin{aligned} 2^3 \times 2^2 &= (2 \times 2 \times 2) \times (2 \times 2) \\ &= 2 \times 2 \times 2 \times 2 \times 2 = 2^5 \end{aligned}$$

$$\text{i.e. } \boxed{2^3 \times 2^2 = 2^{3+2} = 2^5}$$

$$\begin{aligned} \text{Also } \left(\frac{2}{5}\right)^3 \times \left(\frac{2}{5}\right)^4 &= \left(\frac{2}{5} \times \frac{2}{5} \times \frac{2}{5}\right) \times \left(\frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5}\right) \\ &= \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} = \left(\frac{2}{5}\right)^7 \end{aligned}$$

$$\text{ie. } \boxed{\left(\frac{2}{5}\right)^3 \times \left(\frac{2}{5}\right)^4 = \left(\frac{2}{5}\right)^7 = \left(\frac{2}{5}\right)^{3+4}}$$

Hence we deduce that if two rational numbers with same bases and different exponents are multiplied then we just add their exponents keeping the base same.

$$\text{i.e. } \boxed{a^m \times a^n = a^{m+n}}$$

where a is a rational number and m, n are natural numbers.

Example. Simplify using product law.

$$(i) \quad \left(\frac{2}{7}\right)^4 \times \left(\frac{2}{7}\right)^5 \quad (ii) \quad \left(\frac{3}{4}\right)^5 \times \left(\frac{3}{4}\right)^6 \times \left(\frac{2}{5}\right)^4 \times \left(\frac{2}{5}\right)^5$$

Solution: (i) $\left(\frac{2}{7}\right)^4 \times \left(\frac{2}{7}\right)^5$

$$= \left(\frac{2}{7}\right)^{4+5} \text{ (using product law)}$$

$$= \left(\frac{2}{7}\right)^9$$

(ii) $\left(\frac{3}{4}\right)^5 \times \left(\frac{3}{4}\right)^6 \times \left(\frac{2}{5}\right)^4 \times \left(\frac{2}{5}\right)^5$

$$= \left(\frac{3}{4}\right)^{5+6} \times \left(\frac{2}{5}\right)^{4+5} \text{ (using product law)}$$

$$= \left(\frac{3}{4}\right)^{11} \times \left(\frac{2}{5}\right)^9$$

(II) When bases are different but exponents, are same then product law is: $a^n \times b^n = (ab)^n$ where a, b are rational numbers and n is a natural number.

In order to deduce this law, let us consider the following examples.

$$\begin{aligned} 3^4 \times 2^4 &= 3 \times 3 \times 3 \times 3 \times 2 \times 2 \times 2 \times 2 \\ &= (3 \times 2) \times (3 \times 2) \times (3 \times 2) \times (3 \times 2) \\ &= 6 \times 6 \times 6 \times 6 = 6^4 \end{aligned}$$

$$\text{i.e. } 3^4 \times 2^4 = 6^4 = (3 \times 2)^4$$

Similarly $\left(\frac{2}{3}\right)^3 \times \left(\frac{1}{5}\right)^3 = \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{1}{5} \times \frac{1}{5} \times \frac{1}{5}$

$$= \left(\frac{2}{3} \times \frac{1}{5}\right) \times \left(\frac{2}{3} \times \frac{1}{5}\right) \times \left(\frac{2}{3} \times \frac{1}{5}\right)$$

$$= \left(\frac{2}{15}\right) \times \left(\frac{2}{15}\right) \times \left(\frac{2}{15}\right) = \left(\frac{2}{15}\right)^3$$

$$\text{i.e. } \left(\frac{2}{3}\right)^3 \times \left(\frac{1}{5}\right)^3 = \left(\frac{2}{15}\right)^3 = \left(\frac{2}{3} \times \frac{1}{5}\right)^3$$

Hence we deduce that if two rational numbers with the different bases and same exponents are multiplied then multiply their bases keeping the exponent same.

$$\text{i.e. } a^n \times b^n = (ab)^n$$

where a, b are rational numbers and n is a natural number.

Example. Simplify using product law: (i) $\left(\frac{3}{7}\right)^2 \times \left(\frac{4}{5}\right)^2$ (ii) $x^3 \times y^3$

$$(i) \quad \left(\frac{3}{7}\right)^2 \times \left(\frac{4}{5}\right)^2$$

Solution:

$$\begin{aligned} \left(\frac{3}{7}\right)^2 \times \left(\frac{4}{5}\right)^2 &= \left(\frac{3}{7} \times \frac{4}{5}\right)^2 \text{ (Using product law)} \\ &= \left(\frac{12}{35}\right)^2 \end{aligned}$$

$$(ii) \quad x^3 \times y^3$$

$$\begin{aligned} \text{Solution: } &= (x \times x \times x) \times (y \times y \times y) \\ &= (x \times y) \times (x \times y) \times (x \times y) \\ &= (xy)(xy)(xy) = (xy)^3 \\ &\quad \text{(Using product law)} \\ &= x^3 y^3 \end{aligned}$$

EXERCISE 4.2

1. Simplify using product law.

$$(i) \quad \left(\frac{5}{7}\right)^2 \times \left(\frac{5}{7}\right)^4$$

$$(ii) \quad \left(\frac{3}{4}\right)^6 \times \left(\frac{1}{5}\right)^6$$

$$(iii) \quad \left(\frac{1}{7}\right)^5 \times \left(\frac{1}{7}\right)^5 \times \left(-\frac{5}{7}\right)^8 \times \left(-\frac{5}{7}\right)^8 \quad (iv) \quad p^6 \times p^7$$

$$(v) \quad y^5 \times x^{11} \times y^6$$

$$(vi) \quad \left(\frac{4}{3}\right)^5 \times \left(\frac{4}{3}\right)^6 \times \left(\frac{4}{3}\right)^7$$

$$(vii) \quad x^5 \times y^6 \times x^2 \times y^7$$

$$(viii) \quad \left(-\frac{1}{3}\right)^6 \times \left(-\frac{1}{3}\right)^5 \times \left(-\frac{1}{3}\right)^3$$

$$(ix) \quad x^5 \times y^5 \times z^5$$

$$(x) \quad p^2 \times q^7 \times p^5$$

2. Verify that:

$$(i) \quad 2^5 \times 3^5 = 6^5$$

$$(ii) \quad p^7 \times q^7 = (pq)^7$$

$$(iii) \quad \left(\frac{3}{4}\right)^4 \times \left(\frac{3}{4}\right)^8 = \left(\frac{3}{4}\right)^{12}$$

$$(iv) \quad \left(\frac{1}{7}\right)^4 \times \left(\frac{1}{7}\right)^5 = \left(\frac{1}{7}\right)^9$$

$$(v) \quad \left(-\frac{1}{8}\right)^6 \times \left(-\frac{3}{5}\right)^6 = \left(\frac{3}{40}\right)^{16}$$

Quotient law:

I. When bases are same but the exponents are different, then quotient law is: $a^m \div a^n = a^{m-n}$ where a is rational number and m, n are natural numbers.

In order to deduce this law, let us consider the following examples.

$$\begin{aligned} 3^6 \div 3^2 &= \frac{3^6}{3^2} \\ &= \frac{3 \times 3 \times 3 \times 3 \times \cancel{3} \times \cancel{3}}{\cancel{3} \times \cancel{3}} \\ &= 3 \times 3 \times 3 \times 3 = 3^4 \end{aligned}$$

$$\text{i.e. } 3^6 \div 3^2 = 3^4 = 3^{6-2}$$

$$\text{Also } \left(\frac{1}{2}\right)^5 \div \left(\frac{1}{2}\right)^2 = \frac{\left(\frac{1}{2}\right)^5}{\left(\frac{1}{2}\right)^2} = \frac{\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}}{\frac{1}{2} \times \frac{1}{2}} = \left(\frac{1}{2}\right)^3$$

$$\text{i.e. } \left(\frac{1}{2}\right)^5 \div \left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)^{5-2}$$

Hence we deduce that when two rational numbers with same base and different exponents are divided then we just subtract their exponents keeping the base same.

$$\text{i.e. } a^m \div a^n = a^{m-n}$$

where a is a rational number and m, n are natural numbers

Example 2. Simplify using quotient law: $\left(\frac{2}{3}\right)^6 \div \left(\frac{2}{3}\right)^2$

$$\begin{aligned} \text{Solution: } \left(\frac{2}{3}\right)^6 \div \left(\frac{2}{3}\right)^2 &= \left(\frac{2}{3}\right)^{6-2} \quad (\text{using quotient law}) \\ &= \left(\frac{2}{3}\right)^4 \end{aligned}$$

- II. When bases are different but exponents are same, then the quotient law is: $a^n \div b^n = \left(\frac{a}{b}\right)^n$ where a, b are rational numbers with $b \neq 0$ and n is a natural number.

In order to deduce this law, let us consider the following examples.

$$5^2 \div 3^2 = \frac{5^2}{3^2} = \frac{5 \times 5}{3 \times 3}$$

$$= \frac{5}{3} \times \frac{5}{3} = \left(\frac{5}{3}\right)^2$$

$$\text{Also } 7^3 \div 2^3 = \frac{7^3}{2^3} = \frac{7 \times 7 \times 7}{2 \times 2 \times 2}$$

$$= \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} = \left(\frac{7}{2}\right)^3$$

Hence we deduce that if two rational numbers with different bases and same exponents are divided then we just divide their bases keeping the exponent same.

$$\text{i.e. } a^n \div b^n = \left(\frac{a}{b}\right)^n$$

where a, b are rational numbers with $b \neq 0$ and n is a natural number.

Example 1. Simplify using quotient law: $5^7 \div 3^7$

Solution: $5^7 \div 3^7 = \left(\frac{5}{3}\right)^7$

Example 2. Simplify using quotient law $6^5 \div 2^5$

Solution: $6^5 \div 2^5 = \left(\frac{6}{2}\right)^5 = (3)^5$

EXERCISE 4.3

1. Simplify by using quotient law:

- | | | |
|----------------------------|---|---|
| (i) $5^7 \div 5^4$ | (ii) $\left(\frac{5}{7}\right)^5 \div \left(\frac{5}{7}\right)^2$ | (iii) $\left(\frac{1}{2}\right)^6 \div \left(\frac{1}{2}\right)^2$ |
| (iv) $x^{10} \div x^4$ | (v) $(5)^6 \div (3)^6$ | (vi) $(10)^5 \div (2)^5$ |
| (vii) $(-7)^4 \div (-5)^4$ | (viii) $\left(\frac{2}{5}\right)^6 \div \left(\frac{3}{5}\right)^6$ | (ix) $\left(\frac{7}{11}\right)^6 \div \left(\frac{7}{11}\right)^5$ |
| (x) $x^4 \div y^4$ | | |

2. Verify:

- (i) $\left(\frac{2}{9}\right)^5 \div \left(\frac{2}{9}\right)^2 = 2^3 \div 9^3$
- (ii) $(-3)^8 \div (-5)^8 = \left(\frac{3}{5}\right)^{10} \div \left(\frac{3}{5}\right)^2$
- (iii) $\left(\frac{x}{y}\right)^7 \div \left(\frac{x}{y}\right)^5 = x^2 \div y^2$
- (iv) $5^{11} \div 5^8 = (20)^3 \div (4)^3$

Power law:

I. If a is a rational number and m, n are natural numbers, then power law is: $(a^m)^n = a^{mn}$.

In order to deduce this law, let us consider the following examples.

$$\begin{aligned}(7^3)^2 &= (7^3) \times (7^3) \\ &= (7 \times 7 \times 7) \times (7 \times 7 \times 7) = 7^6\end{aligned}$$

i.e. $(7^3)^2 = 7^6 = (7)^{3 \times 2}$

Similarly

$$\begin{aligned}\left\{\left(\frac{11}{12}\right)^2\right\}^4 &= \left(\frac{11}{12}\right)^2 \times \left(\frac{11}{12}\right)^2 \times \left(\frac{11}{12}\right)^2 \times \left(\frac{11}{12}\right)^2 \\ &= \left(\frac{11}{2} \times \frac{11}{2}\right) \times \left(\frac{11}{2} \times \frac{11}{2}\right) \times \left(\frac{11}{2} \times \frac{11}{2}\right) \times \left(\frac{11}{2} \times \frac{11}{2}\right) = \left(\frac{11}{2}\right)^8\end{aligned}$$

i.e. $\left\{\left(\frac{11}{12}\right)^2\right\}^4 = \left(\frac{11}{2}\right)^8 = \left(\frac{11}{2}\right)^{2 \times 4}$

Hence we deduce that if a power of a rational number is raised to another power then we have to only multiply the two exponents keeping the base same.

$$\text{i.e. } (a^m)^n = a^{mn}$$

where a , is a rational number and m, n are natural numbers.

Example. Simplify using power law: (i) $(2^5)^3$ (ii) $\left\{\left(\frac{3}{10}\right)^2\right\}^4$

Solution:

$$(i) \quad (2^5)^3 = (2)^{5 \times 3} = 2^{15}$$

$$(ii) \quad \left\{\left(\frac{3}{10}\right)^2\right\}^4 = \left(\frac{3}{10}\right)^{2 \times 4} = \left(\frac{3}{10}\right)^8$$

Value of exponential expression when exponent is zero.

For zero: $a^0 = 1$, where a is a rational number.

We know that

$$\frac{a^2}{a^2} = \frac{a \times a}{a \times a} = 1$$

$$\text{Also} \quad \frac{a^2}{a^2} = a^{2-2} = a^0$$

$$\text{Hence} \quad a^0 = 1$$

Similarly $5^0 = 1$, $\left(\frac{5}{2}\right)^0 = 1$ and $\left(\frac{2}{3}\right)^0 = 1$

Value of exponential expression when exponent is negative.

For exponent as negative integer: $a^{-m} = \frac{1}{a^m}$

We know that

$$\begin{aligned} \frac{1}{a^2} &= \frac{a^0}{a^2} \\ &= a^{0-2} \\ &= a^{-2} \quad (\text{By quotient law}) \\ \text{i.e. } \frac{1}{a^2} &= a^{-2} \quad \text{or} \quad a^{-2} = \frac{1}{a^2} \end{aligned}$$

Similarly $5^{-3} = \frac{1}{5^3}$ and $6^{-7} = \frac{1}{6^7}$

In general $a^{-n} = \frac{1}{a^n}$

So, $a^{-1} = \frac{1}{a}$

4.2.2 Demonstrate the concept of power of negative integer that is $(-a)^n$ when n is even or odd integer.

Let us consider the following examples

$(-3)^2 = (-3 \times -3) = +9$ (Exponent is even)

$(-3)^3 = (-3) (-3) (-3) = -27$ (Exponent is odd)

$(-3)^4 = (-3) (-3) (-3) (-3) = +81$ (Exponent is even)

$(-3)^5 = (-3) (-3) (-3) (-3) (-3) = -243$ (Exponent is odd)

From above examples we deduce that

$(-a)^n = +a^n$, if n is even

and $(-a)^n = -a^n$, if n is odd

Example. Which of the following are positive or negative?

(i) $(-5)^{11}$ (ii) $(-3)^{20}$ (iii) $(-8)^{113}$ (iv) $(-5)^{40}$

(v) $(-8)^0$ (vi) $(-7)^{-5}$ (vii) $(-110)^{-40}$

Solution:

S.No.	Expression	Positive or Negative	Reason
(i)	$(-5)^{11}$	Negative	The exponent is odd
(ii)	$(-3)^{20}$	Positive	The exponent is even
(iii)	$(-8)^{113}$	Negative	The exponent is odd
(iv)	$(-5)^{40}$	Positive	The exponent is even
(v)	$(-8)^0$	Positive	The exponent is even
(vi)	$(-7)^{-5}$	Negative	The exponent is odd
(vii)	$(-110)^{-40}$	Positive	The exponent is even

EXERCISE 4.4

1. Simplify using power law.

(i) $(5^2)^7$ (ii) $\left\{\left(\frac{1}{3}\right)^6\right\}^8$ (iii) $\{(-3)^2\}^{10}$

(iv) $(x^5)^{20}$ (v) $\left\{\left(\frac{4}{7}\right)^9\right\}^2$ (vi) $\left\{\left(-\frac{3}{4}\right)^{20}\right\}^2$

2. Convert into negative exponent.

(i) $\frac{1}{2^5}$ (ii) $\frac{1}{3^7}$ (iii) $\frac{1}{x^6}$ (iv) $\frac{1}{\left(\frac{4}{5}\right)^6}$

3. Convert into positive exponent.

(i) 5^{-8} (ii) $\left(\frac{1}{3}\right)^{-7}$ (iii) x^{-5} (iv) y^{-7}

4. Identify which of the following is positive or negative.

(i) $(-7)^6$ (ii) $\left(-\frac{11}{2}\right)^{17}$ (iii) $\left(-\frac{1}{5}\right)^{-26}$
(iv) $(-3)^0$ (v) $\left(-\frac{1}{7}\right)^1$ (vi) $\left(-\frac{1}{5}\right)^{-100}$

4.2.3 Apply laws of exponents to evaluate expressions.

Expressions can be simplified easily with the application of laws of exponents as explained in the following examples.

Example 1. Simplify and write in the simple form.

(i) $(2^5 \div 2^3) \times 2^{-2}$ (ii) $\left(\frac{1}{3}\right)^4 \times \left(\frac{1}{3}\right)^2 \times (3)^6$

(iii) $\left(\frac{2}{3}\right)^2 \times \left(\frac{3}{2}\right)^2 + \left(\frac{5}{4}\right)^3 \times \left(\frac{4}{5}\right)^3$

Solution: (i) $(2^5 \div 2^3) \times 2^{-2}$
 $= 2^{5-3} \times 2^{-2} = 2^{5-3-2} = 2^0 = 1$

(ii) $\left(\frac{1}{3}\right)^4 \times \left(\frac{1}{3}\right)^2 \times 3^6$
 $= \left(\frac{1}{3}\right)^{4+2} \times 3^6$
 $= \left(\frac{1}{3}\right)^6 \times 3^6$
 $= \left(\frac{1}{3} \times 3\right)^6$
 $= (1)^6 = 1$

(iii) $\left(\frac{2}{3}\right)^2 \times \left(\frac{3}{2}\right)^2 + \left(\frac{5}{4}\right)^3 \times \left(\frac{4}{5}\right)^3$
 $= \left(\frac{2}{3} \times \frac{3}{2}\right)^2 + \left(\frac{5}{4} \times \frac{4}{5}\right)^3$
 $= \left(\frac{6}{6}\right)^2 + \left(\frac{20}{20}\right)^3 = (1)^2 + (1)^3$
 $= 1 + 1 = 2$

Example 2. Using laws of exponents, find the value of x .

(i) $2^5 \times 2^3 = 2^x$ (ii) $\left(\frac{4}{5}\right)^6 \div \left(\frac{4}{5}\right)^2 = \left(\frac{4}{5}\right)^{2x}$

(iii) $\left\{\left(\frac{3}{4}\right)^2\right\}^x = \left(\frac{3}{4}\right)^8$

Solution:

(i) $2^5 \times 2^3 = 2^x$
 or $2^{5+3} = 2^x$
 $2^8 = 2^x$
 so $8 = x$
 or $\boxed{x = 8}$

(ii) $\left(\frac{4}{5}\right)^6 \div \left(\frac{4}{5}\right)^2 = \left(\frac{4}{5}\right)^{2x}$
 or $\left(\frac{4}{5}\right)^{6-2} = \left(\frac{4}{5}\right)^{2x}$
 or $\left(\frac{4}{5}\right)^4 = \left(\frac{4}{5}\right)^{2x}$
 so $4 = 2x$
 or $\frac{4}{2} = x$ i.e. $\boxed{x = 2}$

(iii) $\left\{\left(\frac{3}{4}\right)^2\right\}^x = \left(\frac{3}{4}\right)^8$
 or $\left(\frac{3}{4}\right)^{2x} = \left(\frac{3}{4}\right)^8$
 so $2x = 8$
 or $x = \frac{8}{2}$
 or $\boxed{x = 4}$

EXERCISE 4.5

1. Simplify and write in simplest form.

(i) $\left(\frac{5}{7}\right)^6 \times \left(\frac{5}{7}\right)^4 \times \left(\frac{7}{5}\right)^{10}$

(ii) $\left\{\left(\frac{2}{3}\right)^2\right\}^3 \div \left(\frac{2}{3}\right)^4$

(iii) $\left(\frac{7}{9}\right)^3 \times \left(\frac{9}{7}\right)^3 + \left(\frac{11}{12}\right)^5 \times \left(\frac{12}{11}\right)^5$

(iv) $x^0 + x^5 \times x^{-5}$

(v) $\left\{\left(\frac{4}{5}\right)^7 \times \left(\frac{4}{5}\right)^2\right\} \div \left(\frac{4}{5}\right)^8$

(vi) $\left\{\left(\frac{7}{8}\right)^6 \div \left(\frac{7}{8}\right)^4\right\} \times \left(\frac{8}{7}\right)^2$

2. Find the value of x in the following.

(i) $3^6 \times 3^4 = 3^x$

(ii) $\left(\frac{11}{9}\right)^8 \div \left(\frac{11}{9}\right)^6 = \left(\frac{11}{9}\right)^{2x}$

(iii) $\left\{\left(\frac{4}{7}\right)^3\right\}^x = \left(\frac{4}{7}\right)^9$

(iv) $\left(\frac{1}{5}\right)^x \times \left(\frac{1}{5}\right)^5 = \left(\frac{1}{5}\right)^{10}$

REVIEW EXERCISE 4

1. Identify the base, exponent and find the value of following.

(i) 3^5 (ii) 7^4 (iii) $\left(\frac{3}{4}\right)^2$ (iv) $\left(\frac{1}{4}\right)^3$ (v) 8^0 (vi) y^0

2. Simplify using laws of exponent.

(i) $\left(-\frac{1}{2}\right)^4 \times \left(-\frac{1}{2}\right)^2$ (ii) $\left(-\frac{4}{5}\right)^2 \times \left(-\frac{4}{5}\right)^2$ (iii) $6^7 \div 6^4$

$$\begin{array}{lll} \text{(iv)} & \left(\frac{2}{7}\right)^5 \div \left(\frac{2}{7}\right)^3 & \text{(v)} \quad (3^2)^5 \quad \text{(vi)} \quad \left\{(-4)^2\right\}^5 \\ \text{(vii)} & (6^0)^2 & \text{(viii)} \quad (5^0)^2 \times (-5) \times (a^0)^2 \end{array}$$

3. Verify:

$$\begin{array}{ll} \text{(i)} & \left(\frac{1}{5}\right)^4 \times \left(\frac{1}{5}\right)^5 = \left(\frac{3}{5}\right)^9 \times \left(\frac{1}{3}\right)^9 \quad \text{(ii)} \quad (6)^4 \div (2)^4 = 3^8 \div 3^4 \\ \text{(iii)} & 6z^0 = 6 \times (z^0)^2 \end{array}$$

4. Which one is positive and which one is negative.

$$\text{(i)} \quad (-2)^7 \quad \text{(ii)} \quad (-5)^0 \quad \text{(iii)} \quad (-3)^8 \quad \text{(iv)} \quad (-6)^{-7} \quad \text{(v)} \quad (-8)^{-10}$$

5. Simplify: (i) $6^2 \div 3^3$ (ii) $5^0 \div 5^{-2} \times 5^2$

$$\text{(iii)} \quad \left\{\left(\frac{2}{5}\right)^2\right\}^3 \div \left(\frac{2}{5}\right)^4 \times \left(\frac{2}{5}\right)^3$$

6. Find the value of x.

$$\begin{array}{ll} \text{(i)} & 25^x = 125 \quad \text{(ii)} \quad \left(\frac{2}{3}\right)^x \times \left(\frac{2}{3}\right)^4 = \left(\frac{2}{3}\right)^{10} \div \left(\frac{2}{3}\right)^2 \\ \text{(iii)} & \left\{\left(\frac{2}{7}\right)^x\right\}^5 = \left(\frac{2}{7}\right)^{10} \end{array}$$

SUMMARY

Base: Base is a number which is to be multiplied by itself according to exponent.

Exponent: Exponent is a number which represents the number of times a base is to be multiplied.

Product law (i) $a^m \times a^n = a^{m+n}$ (ii) $a^n \times b^n = (ab)^n$

where a, b are rational numbers and m, n are natural numbers.

Quotient law (i) $a^m \div a^n = a^{m-n}$ (ii) $a^n \div b^n = \left(\frac{a}{b}\right)^n$

where a, b are rational numbers and m, n are natural numbers.

Power law $(a^m)^n = a^{mn}$

Where a is rational number and m, n are natural number.

Properties (i) $a^0 = 1$ (ii) $a^{-m} = \frac{1}{a^m}$

(iii) $(-a)^n$ is positive if n is even.

(iv) $(-a)^n$ is negative if n is odd.

...

Think Tank: The Most Wonderful Knowledge.

Let some body asks you: Add the consecutive natural numbers from 1 to say 10, ... You can do it within no time as under:

Multiply 10 by its next consecutive natural number

i.e. 11 and divide it by 2.

$$1 + 2 + 3 + \dots + 10 = \frac{10 \times 11}{2} = \frac{\overset{5}{10} \times 11}{\cancel{2}_1} = 55$$

Similarly we can do:

$$1 + 2 + 3 + \dots + 20 = \frac{20 \times 21}{2} = \frac{\overset{10}{20} \times 21}{\cancel{2}_1} = 210$$

$$1 + 2 + 3 + \dots + 25 = \frac{25 \times 26}{2} = \frac{\overset{13}{25} \times \cancel{26}_1}{2} = 325$$

Unit 5 SQUARE ROOT OF POSITIVE NUMBER

5.1 PERFECT SQUARES:

There are certain numbers which can be written in exponential form with exponent 2 for example: $16 = 4 \times 4 = 4^2$.

We read it as '16 is square of 4'. Similarly 9 is square of 3 because $9 = 3 \times 3 = 3^2$

5.1.1 Define a perfect square: In the above examples, 9 and 16 are called perfect squares because these are positive numbers and squares of numbers 3 and 4 respectively.

A positive number which is the square of any number is called a perfect square. Some examples of perfect squares are: 1, 4, 9, 16, 25, 36 etc.

Activity: Complete the following table to find some perfect squares.

Number	Square	Perfect square	Read as
1	$1^2 = 1 \times 1$	1	1 is the square of 1
2	$2^2 = \underline{\hspace{2cm}}$	4	4 is the square of 2
-3	$(-3)^2 = (-3) \times (-3)$	9	<u> </u> is the square of <u> </u>
5	$5^2 = \underline{\hspace{2cm}}$	25	25 is the square of <u> </u>
-6	<u> </u> $= (-6) \times (-6)$	<u> </u>	<u> </u> is the square of (-6)
<u> </u>	$7^2 = 7 \times 7$	<u> </u>	<u> </u> is the square of <u> </u>
8	<u> </u> $= 8 \times 8$	<u> </u>	<u> </u> is the square of <u> </u>
<u> </u>	<u> </u> $= (-9) \times (-9)$	<u> </u>	<u> </u> is the square of <u> </u>
<u> </u>	<u> </u> $= \underline{\hspace{2cm}}$	100	100 is the square of <u> </u>

5.1.2 Test whether a number is a perfect square or not.

We can test a number whether it is a perfect square or not by the following procedure.

1. Find and list its prime factors.